StudySapiens Complete Chapter Notes

Chapter 2 — Bacteria and Viruses

Complete Chapter Notes · Textbook topics converted into board-style long questions, with important conceptual short questions inside every topic · PECTAA 2025–26, Punjab English Medium

This chapter follows the textbook topics 2.1 to 2.8. Each topic is converted into the long questions a board paper can ask from it, and each topic ends with conceptual short questions (why / how / differentiate / justify). Direct definitions are omitted.

Board pattern (Punjab 2026, new syllabus): Chapter 2 carries 1 MCQ, 2 short questions in Q.3, and part (a) of a long question in Q.8. Shorts are 2 marks; longs are 8 marks.

TOPIC 2.1 — STRUCTURE OF BACTERIA

Textbook topic 2.1 · printed pages 33–36

Q.1 Describe the structure of a generalized bacterial cell.

Board long question · 8 marks

Bacteria are a diverse group and all of them have **unicellular prokaryotic organization**, which lacks membrane-bound organelles, including a well-defined nucleus. They have the simplest cellular organization.

Cell Wall

It is a rigid wall around the plasma membrane. The major component is a unique macromolecule called **peptidoglycan** or **murein**. It is composed of long glycan (polysaccharide) chains cross-linked with short peptide fragments. Its amount differs in different bacteria.

The cell wall of **Gram-positive** bacteria contains a **thick layer of peptidoglycan** and has less lipid content. The cell wall of **Gram-negative** bacteria has a **thin layer of peptidoglycan** and an **outer membrane** made of lipopolysaccharides and lipoproteins. This outer membrane makes Gram-negative bacteria resistant to many antibiotics and contains a protein called **porin**, which acts like a pore for specific molecules. Gram-negative bacteria also have more **periplasmic space** than Gram-positive bacteria.

Some bacteria produce a **capsule** outside their cell walls. It is a gelatinous layer and gives sticky character to bacterial colonies.

Cell Membrane

The cell membrane lies just beneath the cell wall. In bacteria that lack a cell wall (e.g., Mycoplasma), it lies outermost. The bacterial cell membrane does not have sterols (e.g., cholesterol). At some points it invaginates and forms vesicles, tubules or lamellae in the cytoplasm; these structures are known as **mesosomes**. They are involved in DNA replication and cell division and also serve as respiratory centres.

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Cytoplasm and Genetic Material

The cytoplasm contains dissolved substances and structures such as the nucleoid, ribosomes and mesosomes. It lacks a cytoskeleton and membrane-bound organelles. Bacterial ribosomes are smaller than eukaryotic ribosomes; each sediments at **70S** (large subunit **50S**, small subunit **30S**). Near the centre of the cytoplasm is an irregular dense area, the **nucleoid**, which contains a **single, circular, double-stranded DNA** without attached histones.

Some bacteria have circular, double-stranded extra-chromosomal DNA molecules called **plasmids**. They are self-replicating and contain genes that enable resistance against unfavourable conditions (e.g., antibiotics). Plasmids also serve as important vectors in genetic engineering.

Q.2 Differentiate between the cell walls of Gram-positive and Gram-negative bacteria. Why is Gram staining important?

Board long question · 5 + 3 marks

Feature	Gram-positive bacteria	Gram-negative bacteria
Peptidoglycan layer	Thick	Thin
Lipid content	Less	More (outer membrane)
Outer membrane	Absent	Present; made of lipopolysaccharides and lipoproteins
Porins	Absent	Present; act as pores for specific molecules
Periplasmic space	Less	More
Gram stain	Retain violet dye; appear purple	Do not retain violet dye; appear in original colour
Antibiotic resistance	Generally less due to no outer membrane	More resistant due to outer membrane

Gram staining, devised by Sir Hans Christian Gram, differentiates bacteria on the basis of cell wall composition. Gram-positive bacteria stain purple because they retain violet dye, while Gram-negative bacteria do not retain the violet dye. This helps in identification and in understanding antibiotic resistance.

Important Conceptual Short Questions — Topic 2.1

S1. Why do Gram-negative bacteria show greater resistance to many antibiotics than Gram-positive bacteria?

Gram-negative bacteria have an outer membrane made of lipopolysaccharides and lipoproteins outside their thin peptidoglycan layer. This outer membrane acts as a barrier and makes them resistant to many antibiotics.

S2. How does the structure of peptidoglycan provide strength to the bacterial cell wall?

Peptidoglycan is composed of long glycan chains cross-linked with short peptide fragments. This cross-linked mesh gives rigidity and structural support to the bacterial cell wall.

S3. Why are mesosomes important in bacteria?

Mesosomes are invaginations of the cell membrane. They are involved in DNA replication and cell division and also serve as respiratory centres, compensating for the absence of membrane-bound organelles.

S4. How do plasmids help bacteria survive unfavourable conditions?

Plasmids are self-replicating extra-chromosomal DNA molecules that carry genes such as antibiotic resistance genes. They can replicate before or after division and help bacteria survive stress.

S5. Why are plasmids used as vectors in genetic engineering?

They are small, circular, self-replicating DNA molecules that can carry selected genes into bacteria for cloning or for the synthesis of specific proteins.

S6. Differentiate between the nucleoid of bacteria and the nucleus of eukaryotes.

The bacterial nucleoid is an irregular dense region containing a single circular double-stranded DNA without histones and without a nuclear membrane. The eukaryotic nucleus has multiple linear chromosomes with histones and is enclosed by a nuclear envelope.

S7. Why do Mycoplasmas lack a cell wall and how does this affect their treatment?

Mycoplasmas naturally lack a cell wall, so antibiotics such as penicillin that inhibit cell wall synthesis are ineffective against them.

S8. How does the capsule benefit bacterial colonies?

The capsule is a gelatinous layer outside the cell wall that gives sticky character to colonies, helping bacteria adhere to surfaces and form colonies.

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TOPIC 2.2 — ENDOSPORE FORMATION IN BACTERIA

Textbook topic 2.2 · printed pages 36–37

Q.3 Explain endospore formation in bacteria. How does it help bacteria withstand unfavourable conditions?

Board long question · 5 + 3 marks

Many bacteria can survive extended periods of harsh conditions by forming specialized **resting cells** called **endospores**. Endospores are **thick-walled** and **metabolically inactive (dormant)**. The process by which bacteria make endospores is called **sporulation**.

1. When a bacterium faces unfavourable conditions, it replicates its DNA.
2. The cell membrane makes a septum to isolate the new DNA and a small portion of cytoplasm.

3. The cell membrane again grows around the new DNA, cytoplasm and septum, so the new DNA is surrounded by two membranes.

4. The DNA of the vegetative cell disintegrates and the whole cell begins to dehydrate.

5. A new peptidoglycan layer forms between the membranes around the separated DNA and cytoplasm, and a spore coat also forms around it.

6. The structure matures into an endospore. The vegetative cell breaks and the endospore is released.

7. The endospore remains dormant until favourable conditions return, then germinates to give rise to a new vegetative cell.

> **Important**
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> Endospores are **not reproductive structures**; they are survival structures. One vegetative cell forms one endospore, which germinates into one vegetative cell.

Important Conceptual Short Questions — Topic 2.2

S1. Why do bacteria form endospores?

Endospores allow bacteria to survive extended periods of harsh conditions. They are thick-walled, metabolically inactive resting cells that remain dormant until favourable conditions return.

S2. How does sporulation differ from binary fission?

Binary fission is a reproductive process producing two identical daughter cells. Sporulation is a survival process in which one vegetative cell forms one dormant endospore, not an increase in number.

S3. Why are endospores resistant to unfavourable conditions?

They are thick-walled, dehydrated and metabolically inactive, with a spore coat and peptidoglycan layer protecting the DNA and cytoplasm.

S4. What happens to the vegetative cell during endospore formation?

The DNA of the vegetative cell disintegrates, the cell dehydrates, and finally the vegetative cell breaks to release the mature endospore.

S5. How does an endospore return to active life?

When favourable conditions return, the endospore germinates and gives rise to a new vegetative cell.

TOPIC 2.3 — MOTILITY IN BACTERIA

Textbook topic 2.3 · printed pages 37

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Q.4 Describe the different patterns of motility in bacteria.

Board long question · 8 marks

- **Flagellar movement (swarming):** Most bacilli and spirilla move by means of flagella. When a bacterial population moves together by flagella, the movement is called **swarming**. Counter-clockwise rotation of the flagellum pushes the cell forward with the flagellum trailing behind.
- **Twitching or crawling:** Used to move over surfaces. It is mediated by **pili**, which bind to a solid surface and retract, pulling the bacterial cell forward.
- **Gliding:** Similar to twitching, but bacteria secrete a **slimy substance** that helps them glide smoothly over solid surfaces.
- **Sliding:** Due to the expansion created by the pushing force of dividing cells.
- **Brownian movement:** Some bacteria (e.g., Streptococcus) that lack flagella or pili move due to random, uncontrolled movements of particles in the fluid.
- **Movement by axial filament:** Spirochaetes have a modified flagellum called an **axial filament**, anchored at one end and running length-wise in the periplasmic space. It consists of two sets of flagella-like fibrils anchored at the two poles and helps in flexing, swimming, creeping and spinning movements.

Important Conceptual Short Questions — Topic 2.3

S1. How does counter-clockwise rotation of flagellum help bacterial movement?

Counter-clockwise rotation pushes the cell forward with the flagellum trailing behind, allowing bacteria to swim in liquid media.

S2. Differentiate between twitching and gliding motility.

Twitching is mediated by pili that bind to a surface and retract, pulling the cell forward. Gliding is similar but involves secretion of a slimy substance that allows smooth movement over surfaces.

S3. Why is Brownian movement not considered true motility?

It is random and uncontrolled movement caused by particles in the fluid striking the bacterium, not by a cellular structure such as flagella or pili.

S4. How does the axial filament help spirochaetes move?

The axial filament lies in the periplasmic space and consists of flagella-like fibrils anchored at the poles. Its contraction produces flexing, swimming, creeping and spinning movements.

S5. How does swarming differ from individual bacterial movement?

Swarming is the coordinated movement of a bacterial population together by means of flagella, whereas individual movement involves a single bacterium swimming by its own flagellum.

TOPIC 2.4 — FLAGELLA

Textbook topic 2.4 · printed pages 38

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Q.5 Describe the arrangement and structure of bacterial flagella. How does it differ from eukaryotic flagella?

Board long question · 5 + 3 marks

Many bacteria have flagella, which enable them to move. The secondary function of flagella is to **detect and respond to chemical signals**.

Arrangement	Description
Atrichous	Bacteria without flagella
Monotrichous	Single polar flagellum
Lophotrichous	Tuft of flagella at one pole
Amphitrichous	Flagella at each of two poles
Peritrichous	Flagella surrounding the whole cell

Structure

The bacterial flagellum is entirely different from the eukaryotic flagellum. It is **not built on the 9+2 pattern of microtubules**; it is composed of **flagellin protein**. It consists of a **basal body**, a **hook** and a **filament**. The basal body lies just beneath the cell membrane and consists of rotating rings — **one pair in Gram-positive** bacteria and **two pairs in Gram-negative** bacteria. The rings anchor the flagellum in the cell membrane and cell wall. The hook is a curved structure connecting the basal body with the filament.

> **Important**
>
> **Pili** are non-helical, filamentous appendages, smaller and thinner than flagella. They are used for attachment to surfaces and are involved in conjugation.

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Important Conceptual Short Questions — Topic 2.4

S1. Why is the bacterial flagellum structurally different from the eukaryotic flagellum?

The bacterial flagellum is made of flagellin protein and lacks the 9+2 microtubule arrangement. Eukaryotic flagella are built on the 9+2 pattern of microtubules.

S2. How does the basal body differ in Gram-positive and Gram-negative bacteria?

The basal body has one pair of rotating rings in Gram-positive bacteria and two pairs in Gram-negative bacteria, reflecting differences in their cell envelope.

S3. Differentiate between monotrichous and peritrichous flagellation.

Monotrichous bacteria have a single polar flagellum, while peritrichous bacteria have flagella surrounding the whole cell.

S4. Why are pili not considered organs of locomotion like flagella?

Pili are mainly used for attachment to surfaces and for conjugation. Twitching motility uses pili, but their primary role is attachment and genetic exchange, not swimming.

S5. How do flagella help bacteria respond to their environment?

Besides movement, flagella detect and respond to chemical signals, helping bacteria move toward favourable conditions.

TOPIC 2.5 — BACTERIA: ECOLOGY AND DIVERSITY

Textbook topic 2.5 · printed pages 39–41

Q.6 Describe the major groups of bacteria and their ecological roles.

Board long question · 8 marks

The fossil record shows that prokaryotes were abundant **3.5 billion years ago** and remained alone on Earth for the next **2 billion years**. Today bacteria are found in water, air, soil, food and in the bodies of animals and plants. They outnumber all eukaryotes and can survive in extreme habitats. Margulis and Schwartz classified prokaryotes into **16 phyla**.

Group	Key features	Ecological role / examples
Omnibacteria	Rigid, rod-shaped, heterotrophic, Gram-negative; usually aerobic; many flagellated; no spores	Pathogens and decomposers; Escherichia coli, vibrios
Cyanobacteria	Photosynthetic; chlorophyll-a, carotenoids, phycobilins; many fix nitrogen in heterocysts	Carbon and nitrogen fixers; increased free oxygen in atmosphere; blooms in polluted water
Mycoplasmas and Spiroplasmas	Lack cell walls; resistant to penicillin	Pathogens; mycoplasma pneumonia; lethal yellowing of coconuts
Spirochaetes	Long spirilla with Gram-negative walls; 2 to >100 flagella	Decomposers and pathogens; Treponema causes syphilis
Pseudomonads	Straight/curved Gram-negative rods; polar flagella; soil and water	Decomposers, plant pathogens, denitrification; Pseudomonas aeruginosa
Actinomycetes	Filamentous growth; resistant spores	Pathogens and nitrogen fixers; dental plaque; leprosy, tuberculosis; source of antibiotics

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Group	Key features	Ecological role / examples
Nitrogen-fixing aerobic bacteria	Gram-negative, mostly flagellated	Free-living and mutualistic nitrogen fixers; Azotobacter converts atmospheric nitrogen into nitrates
Chemosynthetic bacteria	Derive energy from oxidation of inorganic compounds of N, S, Fe	Oxidize nitrogen and sulphur compounds; play role in nitrogen cycle; Nitrosomonas, Nitrobacter

Important Conceptual Short Questions — Topic 2.5

S1. Why are cyanobacteria considered important in the history of Earth?

They played the most important role in increasing free oxygen in Earth's atmosphere through photosynthesis. They also fix atmospheric nitrogen in heterocysts.

S2. How do heterocysts help cyanobacteria?

Heterocysts are special cells in which many cyanobacteria fix atmospheric nitrogen, making them important in the nitrogen cycle.

S3. Why are Mycoplasmas resistant to penicillin?

They lack cell walls. Penicillin and similar antibiotics work by inhibiting cell wall growth, so they have no target in Mycoplasmas.

S4. Differentiate between omnibacteria and pseudomonads.

Omnibacteria are rigid, rod-shaped, heterotrophic Gram-negative bacteria, many of which are pathogens. Pseudomonads are straight or curved Gram-negative rods with polar flagella, found in soil and water, and are important decomposers and denitrifiers.

S5. Why are actinomycetes important in medicine?

Many antibiotics such as tetracycline, chloramphenicol, erythromycin and neomycin were originally derived from actinomycetes. About 150 new antibiotics from actinomycetes are discovered each year.

S6. How do chemosynthetic bacteria contribute to the nitrogen cycle?

They oxidize inorganic nitrogen compounds such as ammonia to nitrite and nitrate, gaining energy for food synthesis. Nitrosomonas and Nitrobacter are examples.

S7. Why can bacteria survive in extreme habitats?

They have diverse nutritional modes, protective structures such as endospores, and metabolic adaptations that allow them to live in water, air, soil, food and inside other organisms.

TOPIC 2.6 — IMPORTANCE OF BACTERIA

Textbook topic 2.6 · printed pages 41–42

Q.7 Describe the ecological and economic importance of bacteria.

Board long question · 8 marks

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Beneficial Bacteria

- **Recyclers of nature:** Bacteria are involved in almost all biogeochemical cycles. Nitrifying bacteria (Nitrosomonas, Nitrobacter, Azotobacter) and denitrifying bacteria (Pseudomonas) complete the nitrogen cycle. Decomposer bacteria decompose dead organic matter and play a key role in the carbon-hydrogen-oxygen cycle. Photosynthetic bacteria such as cyanobacteria increase free oxygen in the atmosphere.
- **Makers of useful products:** Lactobacillus, with yeasts and molds, is used in fermented foods such as cheese, pickles, soy sauce, vinegar, wine and yogurt. Bacteria are used in pharmaceutical and agrochemical industries, antibiotic production and leather processing.
- **Environmental cleaners:** Many bacteria degrade organic compounds and are used in **bioremediation** — removal or degradation of pollutants. They decompose city sewage, digest hydrocarbons in petroleum to clean oil spills, and remediate industrial toxic wastes.
- **Biopesticides:** Bacteria are used in biological pest control, commonly Bacillus thuringiensis, a Gram-positive soil bacterium. These biopesticides are environmentally friendly and have little or no effect on humans, wildlife, pollinators and beneficial insects.
- **Research and technology tools:** Bacteria grow quickly and are easily manipulated. They are used in molecular biology, genetics and biochemistry. Scientists make mutations to study gene function and insert human genes to produce therapeutic proteins such as insulin, growth hormones and antibodies.

Harmful Bacteria

Some bacteria cause diseases in humans, animals and plants, produce toxins, spoil food and damage materials. Their harmful effects must be controlled while their beneficial roles are utilized.

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Important Conceptual Short Questions — Topic 2.6

S1. Why are bacteria called recyclers of nature?

They are involved in almost all biogeochemical cycles, moving essential elements between organisms and the environment. Decomposers break down dead matter, and nitrifying/denitrifying bacteria complete the nitrogen cycle.

S2. How do bacteria help in bioremediation?

Many bacteria degrade organic pollutants. They decompose sewage, digest hydrocarbons in oil spills and break down industrial toxic wastes into harmless products.

S3. Why is Bacillus thuringiensis used as a biopesticide?

It is used in biological pest control and is environmentally friendly, with little or no effect on humans, wildlife, pollinators and most beneficial insects.

S4. How are bacteria used in genetic engineering?

Scientists insert human genes into bacteria to produce therapeutic proteins such as insulin, growth hormones and antibodies. Bacteria grow quickly and are easy to manipulate.

S5. Differentiate between the role of decomposer bacteria and nitrifying bacteria.

Decomposer bacteria break down dead organic matter and recycle carbon, hydrogen and oxygen. Nitrifying bacteria convert nitrogen compounds and complete the nitrogen cycle.

S6. Why are bacteria important in the leather industry?

Commercial preparation of animal skins for leather goods involves the use of bacteria.

TOPIC 2.7 — NORMAL FLORA

Textbook topic 2.7 · printed pages 42–43

Q.8 Explain normal flora and describe the benefits of bacterial flora in humans.

Board long question · 8 marks

In a healthy animal, internal tissues such as blood, brain and muscle are normally free of microorganisms. Surface tissues such as skin and mucous membranes are constantly in contact with the environment and are colonized by certain microbial species. The mixture of organisms regularly found at any anatomical site is referred to as the **normal flora**.

The normal flora of humans consists of bacteria, a few fungi and protists, and some methanogenic archaea. Bacteria are the most numerous and obvious components. The association between humans and their normal flora is **mutualistic**: the flora gets nutrients, a stable environment, constant temperature, protection and transport, while the body gets many benefits.

1. **Synthesis of vitamins:** Bacteria in the alimentary canal produce vitamins and excrete those in excess of their needs. Enteric bacteria secrete **vitamin K** and **vitamin B12**, and lactic acid bacteria produce certain B-vitamins.
2. **Prevent colonization by pathogens:** Normal flora compete with pathogens for attachment sites and nutrients, so pathogens have less chance of entering body tissues.
3. **Inhibit or kill pathogens:** Intestinal bacteria produce substances that inhibit or kill pathogenic bacteria.
4. **Stimulate production of cross-reactive antibodies:** Normal flora behave as antigens and induce an immune response. Low levels of antibodies produced against normal flora cross-react with certain pathogens and prevent infection or invasion.

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Important Conceptual Short Questions — Topic 2.7

S1. Why is the relationship between humans and normal flora called mutualistic?

Normal flora get nutrients, a stable environment, constant temperature, protection and transport, while the human body benefits from vitamin synthesis, protection against pathogens and immune stimulation.

S2. How does normal flora prevent colonization by pathogens?

It competes with pathogens for attachment sites and nutrients, reducing the chance that pathogens enter body tissues.

S3. Why are internal tissues normally free of microorganisms while surface tissues have normal flora?

Internal tissues such as blood, brain and muscle are protected from the environment. Surface tissues such as skin and mucous membranes are constantly exposed and are colonized by microbes.

S4. How does normal flora contribute to immunity?

Normal flora behave as antigens and induce low levels of antibodies that cross-react with certain pathogens, helping prevent infection or invasion.

S5. Which vitamins are produced by enteric bacteria?

Enteric bacteria secrete vitamin K and vitamin B12, which are absorbed from the alimentary canal and distributed in the body.

TOPIC 2.8 — VIRUS

Textbook topic 2.8 · printed pages 43–44

Q.9 Describe the structure of a virus. Why are viruses not considered fully living?

Board long question · 5 + 3 marks

Viruses are extremely small infectious agents and can only be seen under an electron microscope. They range in size from **20 nm** (parvovirus) to **250 nm** (pox viruses). They are 10 to 1000 times smaller than most bacteria and can pass through the pores of filter paper.

- **Genome:** The central core is made of nucleic acid, either **DNA or RNA**.
- **Capsid:** The core is surrounded by a protein coat called the capsid, which gives definite shape to the virus. It is made of protein subunits called **capsomeres**. The number and kind of capsomeres is characteristic of a particular virus.
- **Nucleocapsid:** The central core and capsid together are called the nucleocapsid.
- **Envelope:** In some animal viruses, the nucleocapsid is covered by a lipid-rich membrane derived from the host cell. Non-enveloped viruses are known as **naked viruses**.

Viruses show some characteristics of living things but lack cellular organization and cannot carry out metabolism on their own. They depend on a host cell to

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replicate, so they are not placed in the five kingdoms or three domains.

Q.10 Describe the structure of a bacteriophage (T4 phage).

Board long question · 8 marks

Bacteriophages are a diverse group of viruses that attack bacteria. They are among the most complex viruses. The best known phages of Escherichia coli are **T-phages**.

- **Head:** An elongated pyramidal, hexagonal, prism-shaped structure. Its capsid is made of proteins and the core contains a long double-stranded DNA.
- **Tail:** A straight tail attached to the head, made of an inner core and outer sheath, both of different proteins.
- **Neck:** Attaches the sheath to the head.
- **End plate:** Present on the other side of the sheath.
- **Tail fibres:** Six tail fibres attached to the end plate help the phage attach to the bacterial wall. These structures are also made of proteins.

Important

Bacteriophages are used as carriers in genetic engineering. The gene of interest is inserted into phage DNA, which carries it to the target bacterial cell. When the virus incorporates its DNA into the bacterial chromosome, the gene of interest also becomes part of bacterial DNA.

Q.11 Describe the structure of HIV. Why is it called a retrovirus?

Board long question · 5 + 3 marks

Human Immunodeficiency Virus (HIV) belongs to the group called **retroviruses**. Retroviruses contain **RNA** and their capsids are surrounded by lipid-rich envelopes. The envelope has **glycoprotein spikes**, which help the virus identify and bind to its target. HIV is spherical and about **100 nm** in diameter.

The most distinguishing character of retroviruses is the presence of the enzyme **reverse transcriptase**. This enzyme allows the virus to make DNA from its RNA genome, which is the reverse of the usual transcription process.

Important Conceptual Short Questions — Topic 2.8

S1. Why are viruses called acellular?

They lack cellular organization. They consist of nucleic acid surrounded by a protein coat and cannot carry out metabolism or replicate without a host cell.

S2. How does the capsid determine the shape of a virus?

The capsid is made of protein subunits called capsomeres. The number and kind of capsomeres is characteristic of a particular virus and gives it a definite shape.

S3. Differentiate between enveloped and naked viruses.

Enveloped viruses have a lipid-rich membrane derived from the host cell surrounding the nucleocapsid. Naked viruses lack this envelope and consist only of the nucleocapsid.

S4. Why are bacteriophages used in genetic engineering?

They can carry a gene of interest into a bacterial cell. When phage DNA incorporates into the bacterial chromosome, the inserted gene also becomes part of bacterial DNA.

S5. How do tail fibres help a bacteriophage?

The six tail fibres attached to the end plate help the phage attach to the bacterial wall.

S6. Why is HIV called a retrovirus?

It contains RNA and the enzyme reverse transcriptase, which makes DNA from the RNA genome — the reverse of the usual transcription process.

S7. How do glycoprotein spikes help HIV infect host cells?

They help the virus identify and bind to its target host cell, which is the first step in infection.

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